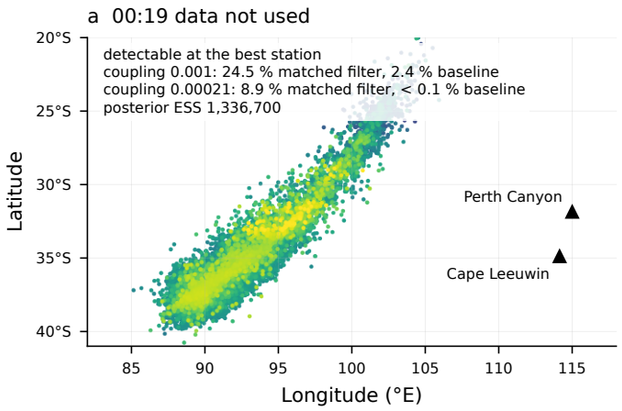
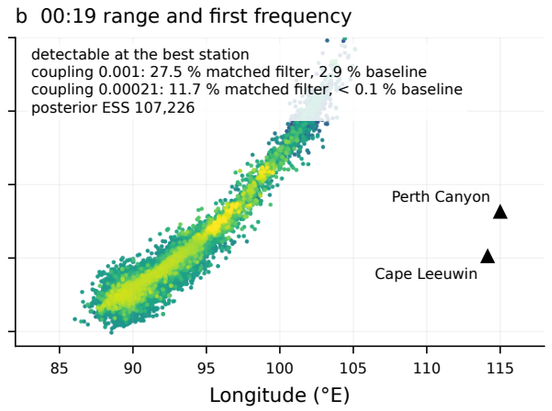
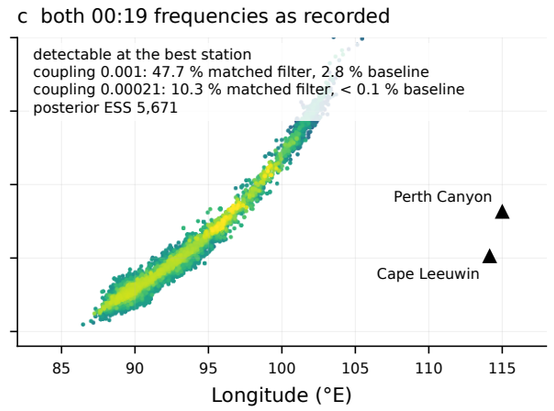
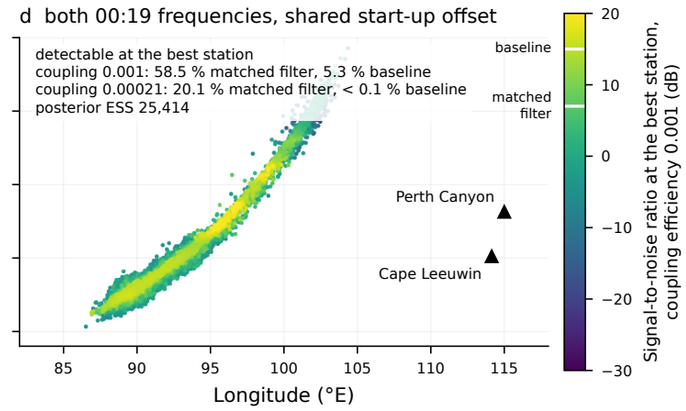
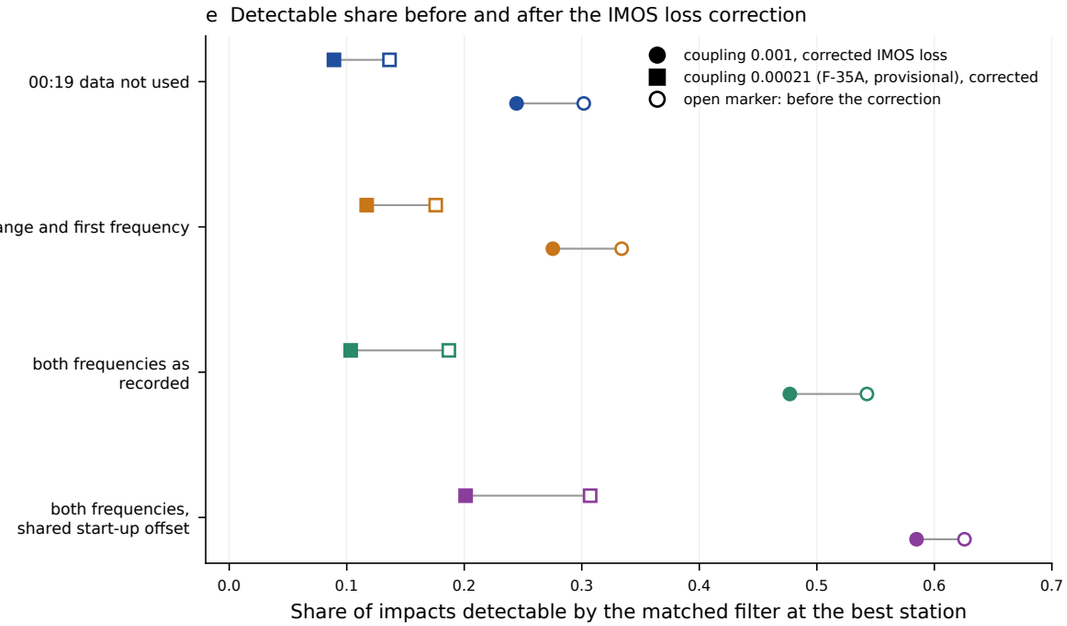
Where the simulated impacts would be heard best: predicted signal-to-noise ratio at the best station, for four uses of the 00:19 data (dive probability 0.5)

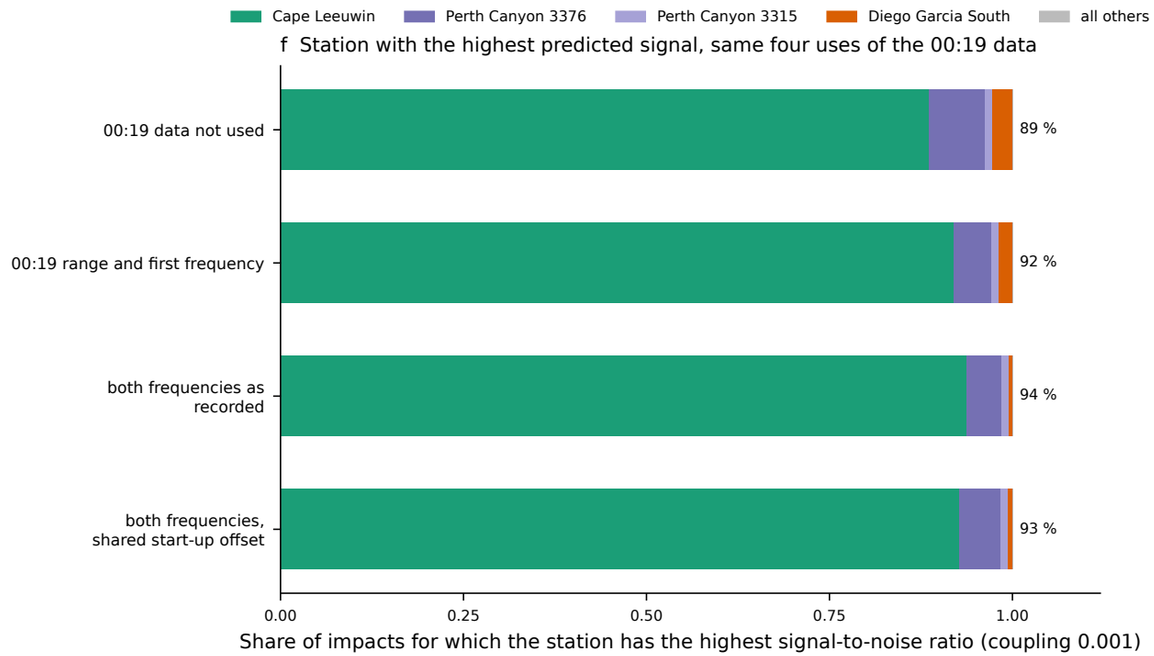












Plain note: Each dot is one likely place where the aircraft hit the ocean, from the end-of-flight run. The colour shows how far the predicted sound at the best station is above or below the noise at that station. Panels a to d show four ways to use the last satellite data at 00:19. Panel e gives the share of impacts that a detector could find at the best station, before and after the correction of the sound loss to the IMOS loggers. Panel f shows which station would hear each impact best. The loudness is not calibrated.

Technical note: Source posterior: end-of-flight run J (74 seeds, 313.6 M rows, four strata pooled by p_core/K) + northern supplement (impacts-ruleB.npz); weighted subsample 2 M draws per 00:19 option at dive probability p = 0.5 (run_j_levels.py, prereg 4277320). Maps: 40,000 of 150,000 draws taken with probability proportional to weight (so dot density = posterior density), drawn in order of SNR (highest last); colour = max over the eight receivers of SNR at eta = 0.001 (5–40 Hz 1 s-equivalent total + 4.8 dB triad gain at IMS − station 5–40 Hz noise median; blocked paths excluded from the max). Shares in the panels and in e/f are weighted over the full 2 M-draw subsample. Posterior ESS = Kish ESS of the full run J mixture at p = 0.5 (run-j-windows/ess.json): D0′ 1.34 M, D2 107 k, D3 5.7 k, D4 25 k; Kish ESS of the 2 M-draw subsample 1.80–1.93 M. IMOS transmission loss: corrected RAM-along-the-arc table imos_tl_ram_arc.csv (imos_ram_grid.py, run_j_levels amendment 2; replaces imos_tl_ramcorr_arc.csv); IMS (H01W/H08S/H08N) loss unchanged. 'Before' = the overnight levels table (imos_tl_ramcorr_arc.csv). Options D1 (R600 BTO only) and D5 (R1200 offset) are not in the run J levels design (run_j_windows OPTIONS = D0′, D2, D3, D4): coverage gap, not estimable from these draws. Latitude span to 20 S (more northerly than the 99th-percentile contour, 27.5 S for D0′). Detectable = SNR >= 7 dB (matchedfilter) / >= 15 dB (baseline). Replaces the 'peak radiated power at impact' six-panel figure (impacts now shown by predicted SNR, not source loudness). PREDICTIVE, NOT CALIBRATED. C1: propagation slope misfit +10.4 / +7.4 dB per octave at H01W / H08S (after the EW0112 source result a source-term caveat; air9 verdict NOT EXPLAINED, gap G23). C2: coupling efficiency eta not calibrated; swept over four decades; F-35A upper anchor eta = 2.1e-4 is provisional (from Brown 2026's quote of Metz 2023; re-derivation from Metz pending). No detection is claimed.